

MODULE 10- INFECTIOUS DISEASES

SECTION 1 Common Vector borne bacterial infections

Relapsing Fever

- **Genus Borrelia has the following medical important spp**
 - Louse-borne relapsing fever (**LBRF**) or **Epidemic relapsing fever**: caused solely by *Borrelia recurrentis*
 - Tick-borne relapsing fever (**TBRF**) - **Endemic relapsing fever**
 - **Borrelia burgdorferi**– Causes **Lyme disease**, another Tick-borne disease.
 - **Borrelia Vincenti**- Together with fusiform bacteria causes ulcerative **gingivostomatitis** (Vincent's angina).
- a- **Epidemic relapsing fever**
 - Caused by Louse borne *Borrelia recurrentis* & its variants
 - *Borrelia* spp. In general, are
 - Endoflagellated spirochetes
 - Don't gram stain because of their thin wall rather they stain with Giemsa
 - Unlike other spirochetes, they can be cultured
 - Either human pathogen or mucus membrane commensal
 - Transmission: Lice bite, crushing lice around wound, etc
 - Bacteremia only in febrile stage of the patient
 - Thus lice get infected if they bite during febrile stage only
 - Repeated bouts of fever (4 days) & no-fever/apyrexia (3-10 days) -----thus a relapsing fever.
 - Reason because of ability of the bacteria to cause antigenic variations that help it escape the immune system.
- b- **Endemic relapsing fever**
 - Tick borne relapsing fever is a **zoonosis**
 - **Clinical features**: The hallmark of both LBRF & TBRF is 2 or more episodes of high fever (usually >39°C), headache, myalgia, epistaxis, etc.
 - The clinical manifestations are also similar for both.
 - **Complications**: Congestive heart failure, Jaundice, Bleeding diathesis, Jarish-Herxheimer reaction: (occurs after 2-4 hrs after administration of antibiotics, during spirochete AL Die-off)
- c- **Diagnosis**:
 1. **Smear**- During febrile stage: blood films reveal large numbers of loosely coiled spirochetes among red cells.
 2. **Animal inoculation**: IP injection of patients' blood into a white mice
 3. **Serology**: complement fixation test (CFT)

Typhus (Rickettsia bacteria)

Disease	Bacterium	Arthropod vector	Primary Host
Epidemic Typhus	R.prowazekii	Human body louse	Humans (Latency in RES)
Endemic Typhus	R.typhi	Rat flea	Rodents
Scrub Typhus	Orientiatsutsugamushi	Chigger Mite larva	Rodents
Rocky mountain spotted fever	R.rickettsii	Ticks	Ticks, dogs, rodents

- General features of Rickettsia spp
 - o Aerobic, gram-negative bacilli (too small to stain well with Gram stain)
 - o Obligate intracellular bacteria (do not make sufficient ATP for independent life)
 - o Non motile & no known toxin or immunopathology
- Common clinical manifestations: - Rash & Vasculitis
- Rash of:
 - o Epidemic & Endemic Typhus → Centrifugally spreading (starts from trunk & spreads to extremities)
 - o Rocky mountain spotted fever → Centripital rash
- Epidemic typhus can also be manifested as
 - o Sylvatic typhus → if caused by bite of flea of flying squirrel
 - o Brill-Zinsser disease → relapse or recrudescence of louse born typhus w/c has been latent in RES of humans
 - No rash is seen here
- Diagnosis
 - o Serological IFA test is most widely used; fourfold increase in titer is diagnostic
 - o Weil-Felix test (cross-reaction of Antigens with Proteus vulgaris) is no longer used
- Treatment: Doxycycline
- Vaccines:
 - R.prowazekii → live & active vaccines
 - Rocky mountain spotted fever → Active vaccine

Ehrlichia, Anaplasma and Coxiella

Coxiella burnetii

- *Areobligate intracellular* rickettsia-like Gram-negative bacteria that cause systemic disease called Q fever (Q = Query).
- grows preferentially in **vacuoles** of eukaryotic cells (infection of macrophages)
- Resistant to dryness - survive for months outside the host cell.
- This may be due to endospore formation.

Q-FEVER

- No arthropod vector
- It stains weakly with the Gram stain.
- **Two structural forms of C. burnetii are recognized**
- **Small cell variants:** resistant to environmental stress. Mature to large cell variants.
- **Large cell variants:** metabolically active form
- **Pathogenesis:** extremely infectious, one organism can cause disease
 - : Human disease is acquired by the respiratory route (inhalation of airborne particles from a contaminated environmental source)
- **Ticks** don't transmit disease to humans, only in animals
- **Clinical manifestation:** incubation period presents with classical symptoms
 - **No rash** with Q fever unlike other rickettsial diseases.
- **Acute diseases:** characteristically flu-like syndrome, **interstitial atypical non-bacterial pneumonia** lasting 5-14 days with a low mortality rate.
- **Chronic diseases:** Endocarditis, granulomatous hepatitis, meningitis, pulmonary disease.
- Most common presentation of chronic Q fever is **subacute endocarditis**.
- **Lab diagnosis: Serology is the most commonly used diagnostic test**
 - : **Weil-Felix test** is negative, but there is a rise in titre of specific Abs to C. burnetii detected by **immune fluorescence assay**.
 - : **Culture**(not commonly performed).
- **Prevention and Control of C. burnetii:** Adequate Pasteurization of milk
 - : Vaccination of human and animal

Bartonella

- It consists of a genus of small Gram-negative aerobic facultative intracellular bacilli bacteria.
- There are three types of medically important bartonella spp
 1. **Bartonella Quintana(trench fever)**
 - **Vector** - human body louse. Organism found in feces
 - The human body louse *Pediculus humanus var corporis* transmits the organisms.
 - **Reservoir - humans**
 - transmission : **Cycle:** human → louse → human
 - **Clinical manifestation:** classical symptoms, Severe pain in the tibia (shin-bone fever),
 - More recently has been associated with **bacteremia, endocarditis, and bacillary angiomatosis**.
 - Mortality rates very low
 - **Diagnosis: Serology** - in reference laboratories
 - : **PCR** – in reference laboratories

:**Bartonella Quintana**- grow in cell-free media, grow on blood agar in 10% CO₂.

2. **Bartonella bacilliformis**

- uses polar flagellum for motility, adheres & invades RBCs.
- **Oroya fever**: is the acute febrile stage
- **Verruga peruana**: is the chronic delayed stage
- Transmitted by: sandfly

3. **Bartonella henselae**

- **B. henselae** is primary (but not the sole) cause of **cat scratch disease & peliosis** of the liver (often called **bacillary peliosis, vascular condition**).
- **B. henselae** is a common cause of culture-negative endocarditis.
- **Cat scratch disease (CSD)**: Also known as cat scratch fever or subacute regional lymphadenitis, is a bacterial infection affecting lymph nodes that drain the sites of inoculation.
- **Cats** transmit the organism when they are parasitized by the fleas - spreads by **fleas**. It is acquired from cat bite or scratch or from cat fleas.
- **Clinical manifestation**: Causes self-limited chronic regional lymphadenopathy primarily in immunosuppressed people.
- **Diagnosis**: serology

4. **Bacillary angiomatosis (BA)**

- Characterized by **vascular skin** lesions proliferative form of Bartonella infection resembling **pyogenic granulomas**.
- It is a systemic illness characterized by lesions similar to those of Kaposi sarcoma in the skin, mucosal surfaces, liver, spleen, respiratory & GI tracts, bone, and brain.
- B. quintana and B. henselae cause BA
- Mostly affect immunocompromised patients particularly with HIV/AIDS.
- **Clinical manifestation**: The 1st sign is appearance of numerous pinpoint purplish to bright red raised spots and nodules up to 10cm on or just under the skin (resemble [Kaposi sarcoma](#)).
 - : Lesions occurring on any part of the body, although they are rarely found on the palms, soles, or in the mouth.
 - : Nodules are firm lumps & do not turn white with firm pressure. If injured, the lesions bleed profusely.
- **Diagnosis**: Biopsy

Plague (Yersinia Pestis)

- Distinguishing feature:
 - Small gram-negative rods with bipolar staining
 - Coagulase positive
- Transmission
 - Wild rodents' flea bite → sylvatic plague
 - Human to human transmission by respiratory droplets
- Pathogenesis & virulence factors
 - Coagulase (from mouth parts of flea); Endotoxin & exotoxin; Antiphagocytic antigens, etc.
- Diseases
 - Bubonic plague
 - Flea bites infected animal and then later uninfected human
 - Symptoms
 - Rapidly increasing fever, Regional buboes, Conjunctivitis
 - Leads to septicemia and death if untreated
 - Pneumonic plague
 - Arises from septic pulmonary emboli in bubonic plague or inhalation of organisms from infected individual
 - Highly contagious
- Diagnosis:
 - Clinical specimens and cultures are hazardous.
 - Serodiagnosis or direct immunofluorescence
 - "Safety pin" staining
- Treatment → Aminoglycosides
- Prevention
 - Animal control; avoid sick and dead animals.
 - Killed vaccine (military)

SECTION 2- Common Vector borne viral infections

Yellow fever (see “zoonosis” part of this module note about this)

Encephalitis viruses

- There are many viruses transmitted between humans that can lead to encephalitis.
- Viruses from the herpes group are familiar
 - o Examples. Chickenpox, cytomegalovirus, and Epstein-Barr virus (the cause of infectious mononucleosis) are herpes viruses.
- All herpes viruses may cause encephalitis occasionally, but herpes simplex virus 1 is the most common culprit.
- The herpes simplex virus 1 causes herpes simplex encephalitis.
 - o The most common cause of non-epidemic (not affecting a large number of people at once) encephalitis in developed countries is the herpes simplex virus.
 - o About 30% of herpes encephalitis infections result from initial contact with the virus.
 - o However, 70% of encephalitis cases occur years after the first infection. This happens when virus particles that have been dormant in the brain since the initial infection become active.
- Ticks or mosquitoes spread certain encephalitis-causing viruses to people.
 - o Insect-transmitted viruses are the most common cause of epidemic viral encephalitis in the U.S.
 - o Viruses that can be transmitted to humans from mosquitoes or ticks are called arboviruses.
- The most common insect-borne encephalitis in the U.S. is St. Louis encephalitis. It can occur in people of all ages, in either rural or urban areas.
 - o The St. Louis encephalitis virus is transmitted by mosquitoes that bite infected birds before biting humans
- California encephalitis is caused by an infection with the LaCrosse virus.
- Western equine encephalitis (WEE)
 - o Horses are the primary carriers of this virus.
 - o Mosquitoes bite infected horses, then pass the virus on to other horses and humans
 - o Most cases are asymptomatic
 - o There is a horse vaccine but none for humans
- Eastern equine encephalitis (EEE) → Most severe of arboviral encephalitis
- The West Nile virus (West Nile Fever) → a flaviviridae that can cause encephalitis
 - o Most cases are asymptomatic
- Other types of insect-transmitted encephalitis are found in different parts of the world.
 - o Encephalitis can also be caused by rabies.
 - o Venezuelan Equine Encephalitis (VEE):- Most cases are asymptomatic
 - o Japanese B Encephalitis (has a vaccine; reservoir-pigs)
- Note EEE, WEE & VEE are antigenically related alpha viruses that belong to Togoviridae.

SECTION 3- Common Vector borne parasitic infections

Malaria

- Part of apicomplexa/sporozoa
 - i.e. apical complex with no locomotory structure
- Host: primary (sexual phase)-female anopheles mosquito; secondary (asexual)-Human
 - Sexual phase → Gametocytic phase
 - Asexual phase → Trophozoites 1st in liver (schizont); then in RBCs (Merozoites)

Species	Disease	Features	Blood smears	Liver stage
P.vivax	Benign tertian	48hr fever spike	Enlarged host cells; amoeboid Trophozoites like ring stage	Persistent hypnozoites thus Relapse
P.ovale	Benign tertian	48hr fever spike	Oval, jagged, infected RBCs	Persistent hypnozoites thus Relapse
P.malariae	Quartan or Malarial	72hr fever spike	Bar and band forms; rosette schizonts	No persistent stage; recrudescence
P.falciparum	Malignant Tertian	Irregular fever spikes; causes cerebral malaria	Multiple ring forms crescent-shaped gametes	No persistent stage; recrudescence

- Recrudescence is a reoccurrence of symptoms from low levels of organisms remaining in red cells.
- Relapse is a return of clinical symptoms from liver stages (hypnozoites).

Leshmaniasis

- Part of hemoflagellates

Species	Disease	Vector/Form of	Reservoirs	Diagnosis
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		transmission		
L.Donovani complex	Visceral leishmaniasis	Sand-fly bite	Urban:humans Rural: rodents & wild animals	Amastigotes in macrophages of Reticuloendothelial system
Many Leishmania spp.	Cutaneous Leishmaniasis (oriental sore, etc.)	Sand-fly bite	Same as above	Amastigotes in macrophages of RES
L.braziliensis complex	MucocutaneousLeishmaniasis	Sand-fly bite	Same as above	Amastigotes in macrophages of RES

- Leishmania all: Intracellular (amastigote form in humans), sandfly vector, stibogluconate as treatment.

Schistosomiasis/Bilharziosis

- A trematode (flake) worm
- General properties of trematodes
 - o Generally flat leaf-shaped
 - o Are hermaphroditic except for Schistosoma, which has separate male and female.
 - o Have operculated eggs (except for Schistosoma), which contaminate water,
 - o The first intermediate hosts are snails
 - o Female within the gynecophoral canal of male

Organism	Common Name	Reservoir	Acquisition	Progression in humans	Snail genus
S.mansoni (with; S.japonicum)	Intestinal schistosomiasis	Cats, dogs, cattle, etc.	Contact with water; skin penetration	Skin penetration (itching) → mature in veins of mesentery → eggs cause granulomas in liver (liver enlargement in chronic cases)	Biomphalaria (mansoni); Oncomelania (japonicum)
S.hematobium	Vesicularschistosomiasis	Primates	Contract with water; skin penetration	Skin penetration (itching) → mature in veins of bladder; Chronic infection has ↑ association with bladder carcinoma	Bulinus

- Life cycle:
 - o Egg having spines → Miracidia (enters snail) → Sporocyst → Cercaria (out of snail & penetrates humans) → schistosomulae → Adult worm
 - o **Egg stage** is the infective stage & finding of eggs in feces/urine is primary way of diagnosis

Lymphatic Filariasis

- Elephantitis; Caused by filarial worms (a type of nematode)
 - o 90% due to Wuchereria Bancrofti; also Brugia Malayi, Brugia Timori
- Vector → Mosquitos (e.g. Aedes, Anopheles); Adult form → **in deep lymphatics (Nocturnal periodicity)**; Reservoir (except W. bancrofti) → Cats, rhesus monkey
 - o DEC provocative test used to force the worm out to superficial vessels
- Manifestation is either
 - o Lymphatic filariasis → manifestation due to the worm or microfilariae (mf)
 - o Occult filariasis → no mf or worm will be present thus manifestation is due to immunohyper-responsiveness to filarial antigens
- Diagnosis:
 - o Giemsa stain to view mf in blood; eosinophilia
 - o Imaging → Filaria dance sign (calcification of node)
 - o Knott's technique → mf in chyluria or hydrocele
- Treatment → DEC, Surgery

Oncocerciasis

- By Onchocerca volvulus
- Causes river blindness, itchy "leopard" rash
- Transmission: vector black fly
 - o Lays eggs around highly flowing watery areas like rivers
- Adults cause onchocercoma (benign nodules) upon entry of skin
 - o Epidemiology onchocercoma
 - If below the waist → Africa
 - On the head & above the waist → South America
- Life cycle limited to superficial structures (e.g. skin, eye, superficial lymphatics) thus usually not seen in blood.
- Microfilariae (mf) → cause oncodermatitis (O.D), lymphadenopathy & eye problems
- Wolbachia → a rickettsia like bacteria that is an endosymbiont of filarial
 - o W/out this bacteria, O. volvulus would produce less inflammation in tissue like eye
- Manifestation
 - o Acute papular O.D → Chronic papular O.D → Lichenified Dermatitis → Atrophy (Skin degeneration) → hypopigmentation (leopard skin)

- Diagnosis
 - o Slit lamp examination of eye; Skin snips show mf; Mazzotti test → provocative diagnostic test (similar to DEC provocative test)
- Treatment: Ivermectin, Surgery

Scabies

- Caused by a mite called *Sarcoptes scabiei*
- Transmission: (infective dose – a single adult female is enough)
 - o Any human contact (primary sexual)
 - o Rarely fomites
- Life cycle:
 - o Egg → 6 legged Larva → 8 legged nymph → 8 legged adults (molting of larva for 10 days gives male & molting for 17 days gives a lifelong fully fertilized female)
- Only adult females burrow skin (to lay eggs) but the male just crawls and feeds
- Pathology: allergic rxn (to anything or fluid or feces related to mite)
- Onset of symptoms of first time infestation with scabies are delayed because the mite delays inflammatory response by down regulating expression of chemokine receptors & surface adhesion molecules
 - o But not so much for the second time since B-cell form memory
- Immune response: Innate (C3 complement); Adaptive (B cell releasing IgE)
- Itching primarily occurs at night
- Extreme manifestation → Crusted (Norwegian) Scabies)
- Diagnosis
 - o Despite seeing some rash patterns it's hard to do it clinically because it mimics other skin problems
 - o Serology also <50% specific
- Treatment: Scabicides like Ivermectin
- “Scabies would never occur if we had B-cells on our skin”

SECTION 4- Zoonosis

Anthrax

- Caused by Bacillus Anthracis (1st discovered by Koch)
 - Large gram +ve Rods in chains! (chains unusual for gram +ve Rods), Obligate Aerobe, Forms SPORES (like clostridium)& Toxins; Catalase +
 - Note its only g +ve in fresh/young culture
 - Single antigenic type of capsule and the capsule is even unique
 - It is unique because it is made of protein (other capsules are made of polysaccharide)- made w/ Poly D. Glutamate
 - Characteristic difference from other non-pathogenic Genus Bacillus:
 - Not motile and has capsule (Vice versa if non-pathogenic)
 - Primary host → Herbivore animals (thus a zoonotic for humans)
 - In humans it's rare, mostly has low risk and **no person to person** transmission.
 - Virulence factors:

Poly D Capsule	Thermophilic AB –toxins (has 3 components)		
Antiphagocytic	Protective Antigen	Edema Factor (EF)	Lethal Factor (LF)
	Mediates the endocytosis of EF & LF	Increased cAMP causing cell to spill water	Exotoxin that cleaves MAP kinase (which is a signal transducer responsible for cell growth), cleaving the signal transduction pathway and causes tissue necrosis

Cutaneous Anthrax	Pulmonary Anthrax (Wool Sorter's Disease)	Enteric Anthrax
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=>Entering of bacteria →Painless Blister →Painless Ulceration→ Painless Black eschar (dry scab surrounded by edema)→ Mostly resolves w/out complication or scar. (may sths spread systematically through lymph nodes) =>Mostly Painless but if felt pain, it's due to lymphatic spread resulting in lymphadenopathy. =>Despite antibiotic treatment the lesion will go through all stages	=> Biphasic illness Initially Flu like with chest pain Acute phase: Pulmonary hemorrhage and shock (post invasion of mediastinal lymph nodes) =>Always fatal (no treatment can stop its invasiveness) => Characteristic symptoms: Mediastinal Adenopathy Mediastinal Widening Pulm. Hemorrhage =>Chest x-ray is primary way of diagnosis here	=>Cause is primarily as a result of lymphatic spread (not ingestion) from a pre-existing lesion
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- Transmission is through spores entering: Cutaneous (most commonly), Inhalation & Ingestion
- Based on the way of transmission and site of infection we have 3 types of Anthrax:
- Diagnosis:
 - o Detection of capsule by **McFadyean's reaction** & Immunofluorescence using Indian ink
 - o Culture: White mucoid colony having **Medusa head like appearance** (Comma shaped)
- Treatment →primarily fluoroquinolone if not doxycycline

Brucellosis

- Caused by a gram –ve bacteria called Brucella spp.
- Animals → Primary hosts (it manifests in them as genital disease); Zoonotic (humans accidental hosts from ingesting/inhaling animal products like dairy products)
 - o Spp. Source of the animal the brucella is transmitted from

B. melitensis (goat)	B. abortus (cow)	B. suis (pig)	B. canis (dog)
Most virulent & prevalent world wide	Most common in USA Moderately virulent	2 nd most common in USA	Moderately virulent Don't cause chronic brucellosis

- **No:** Capsule, Motility, Spore
- Facilitative intracellular, Transmission (direct contact, aerosol, ingestion), Aerobe, Catalase & Oxidase +
- Virulence factors: LPS, poor activation of immune response & inhibition of apoptosis
- Unique ability to invade both phagocytic & non phagocytic cells

- Note: Endoplasmic reticulum of reticulo-endothelial organs (Bone marrow, liver, spleen & LNs) is the primary site of replication & spread thus it's our primary site to take a specimen for diagnosis
- Characteristic symptom → Undulating fever
 - Note one of signs of chronic infection is Osteomyelitis
- **Potential Biological weapon (Biosafety level 3) w/out any vaccine.**
- Diagnosis: primarily serology; We can also do a culture
- Treatment: primarily Doxycycline with Rifampin.

Rabies

- (-) sense ssRNA
 - Enveloped
 - Capsule = bullet-shaped
 - Helical nucleocapsid
- Clinical Manifestations
 - In general:
 - Tingling, muscle spasms
 - Salivary glands – inc. saliva production, dysphagia (throat/larynx mm spasms)
 - “Foaming at the mouth”
 - Spread to brain – high fever, encephalitis, neuronal death (fatal)
 - We can classify the above manifestations in to two clinical forms:
 - Paralytic (dumb phase): symptoms are limited to muscles & site of bite
 - Encephalitic (80%) = Paralytic phase + Other spread related characteristics
 - Hydrophobia & Aerophobia are characteristic of this phase
 - In summary:
 - Exposure → prodromal symptoms → 1st neurological sign → Comma → Death
- Transmission:
 - Zoonotic – bats most common carrier (also foxes, skunks, rodents)
 - Also termed as Bite of Rabid animals
 - Both contamination of pre-existing wound & transmission across intact mucus membrane are possible but are unlikely.
- Pathogenesis + Virulence Factors
 - Glycoprotein binds nicotinic ACh receptors in post-synaptic membrane of NMJ
 - Replicates in motor neurons
 - Incubation: weeks-months (2wk-1yr for animals; 1-2month for humans), then symptoms
 - Retrograde travel along peripheral neurons to Dorsal Root Ganglion

- Antidote: ASAP, prior to symptoms give vaccine w/c can be either:
 - o Passive immunization w/ human rabies Ig (pre-formed)
 - **Never give** for post exposure patients & category I rabies patients.
 - o Active immunization – killed vaccine
- Vaccinate anyone exposed to bat even if you can't see bite marks (can be small!)
- **A disease of the Asians (based on its epidemiology)**

Hemorrhagic fever viruses

- Viral hemorrhagic fever is a general term for a severe illness, sometimes associated with bleeding, that may be caused by a number of viruses.
- The term is usually applied to disease caused by
 - o **Arenaviridae**
 - o **Bunyaviridae**
 - o **Filoviridae**
 - o **Flaviviridae** (yellow fever, dengue, Omsk hemorrhagic fever, Kyasanur forest disease)
- Common feature of all:
 - o All are enveloped RNA Viruses
 - o All have animal or insect reservoirs as primary host
 - Humans aren't the primary reservoirs or hosts
 - o All are geographically restricted to the areas where their host species live.
 - o With a few noteworthy exceptions, there is no cure or established drug treatment for VHF
 - Animals control = disease control
- Hemorrhagic fever isn't always caused by viruses. E.g. Scrub typhus
- Arenaviridae: (E.g. Lassa Virus, lymphocytic choriomeningitis virus/LCV)
 - o Primarily (-) sense ssRNA w/c is segmented (2 segments)
 - BUT capacity to encode both positively AND negatively (**ambisense**)
 - o Rodent born (ROBO) virus
 - o Characteristics electron microscopic diagnosis b/c of its Sandy/Grainy Appearance
- Bunyaviridae
 - o -ve sense ssRNA virus w/c is segmented (3 segments)
 - o 4 genera

Bunya virus	Nairo Virus	Hanta Virus/Korean Hemorrhagic virus	Plebo virus
Includes California encephalitis & Rift valley virus	Includes Crimean Congo hemorrhagic virus		

- o Hanta virus: a rodent born (ROBO) virus

- Clinically: Mostly lifelong infection w/out any deleterious effect
 - But signs are present they primarily Pulmonary edema via capillary leak, pre-renal azotemia (capillary leak, fluid loss)
 - Rift Valley fever:
 - A zoonotic virus transmitted by mosquito & sand-fly vectors
 - No direct human to human transmission
 - 3 Clinical forms: Subclinical form (in most cases); Hemorrhagic form; Meningoencephalitis
 - **Retinitis** is one of the most common complication.
 - Crimean Congo H.V. is the only one of this family that can be transmitted direct human to human contact
- Filoviridae (genus Ebola & Marburg):
 - –ve sense ssRNA virus
 - Its genera are: Antigenically different but morphologically identical
 - Clinical manifestations:
 - Hemorrhagic fever, petechial rash (not always present) – start days to weeks after infection
 - End-organ failure – Cause of DEATH
 - Occurs because of Severe blood loss, hypovolemic shock
 - Transmission: Direct contact w/animal (specially from fruit bat & other primates)
 - Direct human to human transmission is possible despite being zoonotic
 - Pathogenesis + Virulence
 - 1st attack macrophages & lymphocytes to be disseminated
 - After dissemination, they increase pro-inflammatory cytokines to make pores in blood vessels → Hemorrhage
 - Out of Ebola strains:
 - Reston strain – disease in non-human primates only
 - Mutated Zaire strain – the recently seen type that caused was pandemic/epidemic.
 - Note: If a person dies of these, it means that no significant immune response has developed.
- Flaviviridae (E.g. yellow fever, Dengue)
 - +ve sense ssRNA virus
 - Yellow Fever
 - 1 serotype, > or = 7 genotype; only found in Africa & South America
 - Transmission is through Aedes Mosquito (thus an ARBO virus)
 - Mosquito bite on skin → skin is site of viral replication → Spread through nodes to Reticuloendothelial system (RES), Kidney & Myocardium (cause of shock)
 - Clinical manifestations:

- Hemorrhage & Viremia during early infection
 - Result in Jaundice, Backache, bloody diarrhea/vomiting
- Necrosis & lesion due to propagation in an organ.
 - Hepatorenal failure is a sign of severe stage.
- Diagnosis:
 - Viral isolation by taking specimens from:
 - Pre mortem: blood only in the first 4 days post infection
 - Post mortem: intracerebral tissue
 - Serology: - primarily IgM Abs in the 1st week of illness.
- Treatment: Live attenuated vaccine; but no anti-viral drug
 - Vaccine not for infants, pregnant, immunodeficients& egg-allergy having people
- 2 epidemiologic cycles
 - Jungle/Rural/Sylvatic Y.F -----here we can say that Y.F is Zoonotic too.
 - Monkey (primary host) → Mosquito → primarily Monkey (but sometimes Human)
 - Urban/Epidemic
 - Human (primary host) → Mosquito → Human
- Dengue Fever (Break Bone Fever)
 - Also an ARBO virus (transmission by Aedes mosquito)
 - 4 serotypes of those type 2 is clinically important because it
 - Infects bone marrow, weakening it and resulting in Break bone fever
 - Also causes Renal failure
 - Most severe form is Dengue Hemorrhagic Fever (primarily seen in children)
 - Pathogenesis of this requires the presence of pre-existing non-neutralizing antibody of different dengue fever serotype. (thus highly related with immunopathology)
 - Isolation of virus is difficult
 - No vaccine or antiviral drug.

Toxoplasmosis

- Caused by **coccidia** called Toxoplasmosis gondii (An obligate intracellular parasite).
 - Three parasitic stages: oocyst, tachyzoite, bradyzoite (are asexual stages)
- Chiefly an infection of animals
 - Felines like Cats → most common definitive hosts (Since sexual stage with excretion of oocysts occurs here)

- Opportunistic infection to humans (Note that its benign/asymptomatic in immunocompetent)
 - 1/3 of world population is infected but problematic in pregnancy and AIDS patients
 - Humans are also referred to as intermediate host (since no sexual stage occurs here)
- “Behavior changer” → research show that it’s due to epigenetic remodeling
 - E.g. Infected Rats/mice – unafraid of cats; Infected women – more likely to cheat their husband; Infected men – aggressive
- Humans get infected through:
 - Primarily: Feco-orally = oocyst; undercooked infected meat = toxoplasma cyst; congenital
 - Additional ways: blood transfusion
- In human, life cycle
 - Bradizites in tissue cyst or Sporozoites in oocysts released following ingestion
 - These Enter small intestinal epithelium, transform to tachyzoites and multiply in all tissue cells **except RBCs** because they attack only nucleated cell types. (Active multiplication = **Acute toxoplasmosis**)
 - >80% of Acute toxoplasmosis is asymptomatic
 - Tissue cysts form mainly in CNS and Muscle – life time (Latency)
 - Reactivation tissue cyst occurs following immune impairment (thus opportunist)
- In immunocompromised patients (like AIDS patients):
 - Primarily manifested as Toxoplasmosis Encephalitis (> **95% TE** – due to reactivation of tissue bradyzoites)
 - In fact, it is the **Leading cause of CNS disease in AIDS patients**
 - Other characteristic manifestations: Chorioretinitis, Pneumonitis (**often confused with PCP**), Multi-organ involvement with respiratory failure & shock
- Most infected mothers remain asymptomatic
- Congenital toxoplasmosis → Most severe form of toxoplasmosis is seen here
 - Infection can be acquired either due to
 - Active parasitaemia during pregnancy or
 - Reactivated in immunocompromised mothers
 - Better for the fetus b/c reinfected mothers usually produce IgG that can/may better equip the fetus against congenital toxoplasmosis.
 - The risk of maternal-fetal transmission increases as the pregnancy proceeds but the consequences become less severe and vice versa
 - Easy way to remember this is based on stage of pregnancy
 - **Early: low transmission, but severe disease**
 - **Late: high transmission, more benign symptoms**
 - Despite being most severe form, it most prevalent as being asymptomatic at birth

- Characteristic manifestations:
 - A tetrad of chorioretinitis, hydrocephalus, epilepsy & Intracranial calcification.
 - If worse → Abortion
- In summary based on prevalence is:
 - Subclinical > Tetrad of symptoms > Abortion
- Diagnosis → primarily serology
- Treatment:
 - Trimethoprim-sulfamethoxazole (TMP-SMX)/pyrimethamine plus sulfadiazine
 - To prevent intrauterine transmission → **Spiramycin**

Hydatidosis (Dog tape worm)

Cestode	Transmission	Disease	Diagnoses	Treatment	Definite & Intermediate hosts
Echinococcus granulosus	Ingestion of eggs	Hydatid cyst disease/ liver & lung where cysts containing brood capsules develop	Imaging; Serology	Surgery (best), Albendazole (best works during intestinal phase)	IH: herbivores; rarely humans DH: Carnivores in sheep-raising areas
Echinococcus multilocularis	Ingestion of eggs	Alveolar hydatid cysts	Imaging; Serology	Same as above	IH: rodents DH: canines and cats

- Humans – accidental hosts; Also there is no herbivore animal to human transmission
- Adult form in Definite host's intestine; Larval form (hydatid cyst) → liver & lung of Intermediate host
- Abdominal symptoms in humans (other than liver) are due to hatching of oncosphere & penetrating intestine to reach liver or lung.
- Cause of death in humans is usually due to heart problems.

SECTION 5- Systemic/Deep Mycosis

- All are dimorphic
- **Blastomycosis (Blastomyces dermatitidis)**
 - Environmental form: hyphae with nondescript conidia (i.e., no fancy arrangements)
 - Association not definitively known, appears to be associated with rotting wood such as beaver dams
 - Tissue form: broad-based budding yeasts and a double refractile cell wall (not capsule)
 - Blastomycosis
 - Acute and chronic pulmonary disease
 - Considered less likely to self-resolve than Histoplasma or Coccidioides, so many physicians will treat even acute infections.
 - Ability to disseminate.
 - Treatment: Itraconazole for mild, amphotericin B for severe
- **Coccidioidomycosis (Coccidioides immitis)**
 - Environmental form: hyphae breaking up into arthroconidia found in desert sand
 - Arthroconidia are inhaled, round up, and enlarged, becoming spherules inside which the cytoplasm walls off, forming endospores.
 - Tissue form: spherules with endospores
 - Disease: Valley fever (asymptomatic to self-resolving pneumonia)
 - Desert bumps (erythema nodosum) and arthritis are generally good prognostic signs.

- Pulmonary lesions have a tendency to calcify as they heal.
 - Systemic infections are a problem in AIDS and immunocompromised patients in endemic region (meningitis, mucocutaneous lesions).
 - Tendency to disseminate in third trimester of pregnancy.
- Treatment: Azoles for mild to moderate (itraconazole, etc.), amphotericin B for severe
- **Histoplasmosis (*Histoplasma capsulatum*)**
 - Environmental form: hyphae with microconidia and tuberculate macroconidia
 - Found in soil (dust) enriched with bird or bat feces
 - Risk factors for humans: Spelunking (cave exploring), cleaning chicken coops, or bulldozing starling roosts
 - Tissue form: small intracellular yeasts with narrow neck on bud; no capsule
 - Facultative intracellular parasite found in reticuloendothelial (RES) cells
 - Disease: Fungus flu (a pneumonia)
 - Asymptomatic or acute (but self-resolving) pneumonia with flu-like symptomatology
 - Hepatosplenomegaly may be present even in acute pulmonary infections (facultative intracellular RES)
 - Very common in summer in endemic areas: children or newcomers (80% of adults are skin-test positive in some areas)
 - Lesions have a tendency to calcify as they heal.
 - Relapse potential increases with T cell immunosuppression.
 - Disseminated infections: Mucocutaneous lesions are common; also common in AIDS patients in endemic area.
 - Treatment: Itraconazole for mild, amphotericin B for severe.
- **Paracoccidioidomycosis (*Paracoccidioides brasiliensis*)**
 - characterized by formation of ulcers on the mucosal surfaces of the mouth that spread to lips, nose, and cheeks, by great enlargement of lymph nodes esp. of the throat and chest, and by involvement of the gastrointestinal tract
 - The infection is more common in males
 - It has latent form

SECTION 6- Mosquitoes, Sand-fly & Snails

Mosquitoes (Culicidae family)

Structure of Body

Head	Thorax	Abdomen
Pair of compound eyes Long Probosis for biting A pair of palpi& antennae	Dorsal wings & Ventral legs	10 Segments (the last 2 segments are modified to form external genitalia) For digestion & Reproduction

Complete Metamorphosis (4 stages)			
Egg	Larva (Wiggler)	Pupa (tumbler)	Adult
Laid on water surface (can't withstand dessication)	3 body segments Except anopheles all others have siphon (air tube) thus anopheles lie more horizontal while the rest are downwards at an angle	2 body segments: Cephalothorax & Abdomen Inactive stage	Only female bites for blood meal (for dev.t of egg) Venation & Scaling on the wing (characteristic feature)

- Anopheles Mosquito
 - o Only some are vectors of malaria
 - o Transmit → Malaria, Filariasis, Arbovirus

- Prefer:
 - Clean water for breeding; feeding at evening/night either indoor (endophilic) or outdoors (exophilic)
- Eggs: boat shaped, lateral floats, laid singly
- Different from other mosquitoes by
 - Palps are as long as the proboscis
 - Black & white scales on the wings
 - Resting position of adult → at an angle to the surface (head & body are in 1 axis)
 - Resting position of larva → horizontal to the water surface
- Culicinae Mosquitos (Aedes, Culex, Mansonia)
 - Unlike anopheles
 - Eggs have no float
 - Larva have siphon (thus at angle to the water surface)
 - Exception: Mansonia larva inserts its siphon into aquatic plant & remains submerged
 - Adults are parallel to the surface (head & body are in 2 axis)
 - Aedes Mosquito
 - Tiger mosquito (white stripes on black body & leg)
 - Breed in artificially collected water
 - Vectors of Arbovirus & sfilariasis
 - Eggs are laid singly; Siphon is short, stout & has a single pair of hair tuft
 - Culex mosquito
 - Nuisance mosquito that prefer dirty water for breeding
 - Eggs are laid in rafts; Siphon is long, slender & has multiple pairs of hair tufts
 - Mansonia
 - Big with speckling on wings & legs
 - Breed at ponds with aquatic plants
- For exophilic/exophagic mosquitoes primary way of control is source reduction where as for endophilic/endophagic mosquitoes it's utilization of chemical insecticides.

Sand fly (Phlebotominae subfamily)

- Phlebotomus (old world spp); Lutzomyia (new world spp)
- Sandy colored; complete metamorphosis; day or night exo/endo-phagic insects.
- Vectors of Sandfly fever, Leishmania, Bartonella
- Females need blood to mature the eggs
- Adults are easily recognized by
 - Hairy moth like appearance
 - Conspicuous pair of black eyes

- Relatively long legs
- When resting, wings are held vertically up over the body

Complete Metamorphosis (4 stages)			
Egg	Larva	Pupa	Adult
Laid on ground or holes (e.g. tree holes)	Primarily in hyrax “Matchstick” hairs	Coma shaped	Bite nocturnally but sometimes also during overcast and in darkened rooms Weak fliers (hopping type of flight)

- They modulate macrophage function to inhibit
 - Killing of leishmania
 - NO production
- They are highly susceptible to insecticides like DDT.

Snails(Planorbidae family)

- Spp. Grouping is based on shape, diameter & number of whorls on the shell
- 2 main groups
 - Aquatic (only under water) -----additionally these are hermaphroditic snails
 - Biomphalaria (S.mansonia)
 - Bulinus (S.hematobium or S. intercalatum)
 - Amphibians (both water & land) -----are separate sex having (non-hermaphroditic)
 - Oncomelania (S.japonicum)
- Behavior of snails
 - Hibernate for cold & drought
 - Don't see & hear but have sense smell that helps in directing their locomotion
 - Nocturnal
 - They die if they consume salt or sugar.
- They cause schistosomiasis, food borne trematode infections (e.g. fascioliasis, paragonimas, etc.)

Differences between Biomphalaria&Bulinus	
Biomphalaria	Bulinus
Flattened with 3-7 whorls on their shells	Left handed opening shells

- Control of snails
 - Biologically by carnivorous snails (e.g. Maris)

- Environmental management
- Chemical control (Molluscicides)

Section 7- Pathology of Selected Fungal Diseases

Fungi are eukaryotes with a rigid, chitin cell wall and plasma membrane with ergosterol in place of cholesterol. They form either sexual or asexual/**conidia** (more common) spores. Conidia as well as other volatile fungi can be inhaled, leading to infections. Adherence factors and invasive factors increase the virulence of the fungi.

Adhesion is mediated by fungal adhesins, on the cell wall, and host cell receptors.

- Eg. *Candida albicans* have mannoprotein components extending from their cell wall that function as adhesins and fibronectins.

Fungi can grow either in the hyphae form or as rounded yeast cells while others can be dimorphic (occurring as both forms at different temperatures). Their hyphae structure, especially, enables them to invade host tissue easily once the species has been introduced in the area while the rounded cells have increased adherence ability. Dimorphic fungi have increased virulence as they have both advantages.

Fungi don't produce exotoxins in vivo hence most of the tissue injury and necrosis they cause is due to inflammatory and immunologic responses. However, some fungi like *Aspergillus* produce exotoxins/ mycotoxins in the environment.

Candida

C. albicans exists in the rounded yeast form as a normal flora (oropharynx, GIT, female genital tract); switch into the hyphae form indicates invasion and enhances the pathogenicity of the species.

The germ tube and hyphae of C. albicans have **Hyphal Wall Protein (Hwp)** which can easily attach to epithelial cells due to its similarity with mammalian keratinocyte transaminase. **Mannoproteins** also have the ability to bind ECM components due to their structural similarity with vertebral integrins. In addition, hyphae also secrete proteinases and phospholipases that digest epithelial cells promoting invasion.

C. albicans has protein surface receptor that binds C3 proteins preventing opsonization.

Section 8- Pathology of Common Viral Diseases

Arboviral Diseases

Arthropod borne viral diseases are classified into three based on their tropism: CNS-related, visceral-related and hemorrhagic fever.

Mode of infection is similar in most cases:

Human infected from bite of an infected arthropod → Viremia, amplified by extensive virus replication in the reticuloendothelial system and vascular endothelium → Virus becomes localized in various target organs, depending on its tropism → Virus produces cell necrosis with resultant inflammation which leads to fever in nearly all infections

Togaviridae and Flaviviridae Encephalitis

After subcutaneous inoculation, the virus replicates in lymphoid cells in nearby lymph nodes and myeloid cells as well as vascular endothelial cells. They then get into the blood stream, cross the BBB via unknown mechanisms eventually causing encephalitis.

Flaviviridae and Bunyaviridae Hemorrhagic Fever

Yellow Fever

In yellow fever, the liver is the primary target organ. Pathologic findings include hyaline necrosis of hepatocytes, which produces cytoplasmic eosinophilic masses called **Councilman bodies**. Degenerative changes in the renal tubules and myocardium may also be seen, as may

microscopic hemorrhages throughout the brain. Hemorrhage is a major feature of yellow fever, largely because of the lack of liver-produced clotting factors as a result of liver necrosis.

Dengue Fever

Uncomplicated dengue fever is related with changes in the small dermal blood vessels. These alterations include endothelial cellswelling and perivascular edema with mononuclear cell infiltration. The more severe form, as in dengue hemorrhagic fever which is often complicated by shock, is characterized by perivascular edema and widespread effusions into serous cavities such as the pleura. This is believed to be due to increased vascular permeability and disseminated intravascular coagulation (due to inflammation).

Zoonotic Viral Diseases

Rabies

Inoculation (from either bite or infected bat droppings) → Virus replicates in muscles near inoculation site → Virus reaches peripheral nerves and climbs up to the CNS (replicates only in gray matter) Passes along autonomic nerves to reach other tissues (salivary glands,...) Once in the CNS, the virus causes plasma cell and lymphocyte infiltration which leads to nerve cell destruction. Throughout the brain, eosinophilic cytoplasmic inclusions are formed, called the Negri, and are particularly abundant in the hippocampus, cerebral cortex and the dorsal spinal ganglia.

Section 9- Pathology of Some Parasitic Diseases

Malaria

Sporozoites from the mosquito are injected into the human and circulate in the peripheral blood Within 1 hour they attach to and invade hepatocytes → The sporozoites then initiate exoerythrocytic schizogony, producing merozoites One to two weeks later, the infected hepatocytes rupture, releasing merozoites into the general circulation The merozoite then attaches to a specific receptor on the RBC surface After attachment, the merozoite invaginates the cell membrane and is slowly endocytosed The merozoites then begin to multiply until the erythrocytes rupture releasing merozoites which can infect new RBC's.

*The sporozoites ability to identify and invade hepatocytes is mediated by a ligand present in the sporozoites' outer protein coat (circumsporozoite protein).

Erythrocytic invasion by the plasmodium is what causes fever, anemia, circulatory changes, and other immunopathologic phenomena characteristic of malaria patients.

Fever: Related with RBC rupture in addition to possible releases of either parasite-derived pyrogens are released at the time of sporulation or release of interleukin-1 (IL-1) and/or tumor necrosis factor (TNF) from macrophages involved in the ingestion of parasitic or erythrocytic debris.

Anemia: Depression of marrow function, sequestration of erythrocytes within the enlarging spleen, and accelerated clearance of non-parasitized cells all appear to contribute.

In addition, elevated levels of IL-1 and TNF are consistently found in patients with malaria (probably released at the time of sporulation) contribute to other symptoms of malaria including high fever due to vasodilation.

Trypanosomiasis

After inoculation of the parasite from a bite from a TseTse fly, a localized inflammatory lesion appears due to multiplication of the trypomastigotes. The development of this chancre is followed by the spread of the organism through lymphatic channels to the bloodstream, inducing lymphadenitis.

Trypanosomes localize in the small blood vessels of the heart and CNS resulting in endothelial proliferation and a perivascular infiltration of plasma cells and lymphocytes. In the brain, hemorrhage and a demyelinating panencephalitis may follow.

Leishmaniasis

After the host is bitten by an infected sandfly, the parasites disseminate in the bloodstream and are taken up by the macrophages of the spleen, liver, bone marrow, lymph nodes, skin, and small intestine. Histocytic proliferation in these organs produces enlargement with atrophy or replacement of the normal tissue and consequent symptoms of the infection.

Toxoplasmosis

If it's a primary infection, the proliferation of trophozoites results in the death of involved host cells, stimulation of a mononuclear inflammatory reaction, and a parasite-specific secretory IgA response. The inflammatory reaction is responsible for most symptoms.

However, acute infection is rapidly controlled with the development of humoral and cellular immunity in patients who had previous exposure to the disease.

Section 10- Drug Resistance

One of the recent threats in the medical world is the rise of drug resistant strains. There is a constant increase in the number of mortalities from diseases we once thought had been solved. Some of these strains include MRSA (Methicilin Resistant Staphylococcus Aureus), VRE

(Vancomycin Resistant Enterococcus), MDRS.typhi (Multi-Drug Resistant Salmonella Typhi) and so on...

Methods of Resistance

- ♣ Increasing impermeability to the drug
- ♣ Altering the drug's target
- ♣ Enzymatic drug modifications
- ♣ Increasing the efflux of the drug (decreasing intracellular concentration of the drug)

Methods of Resistance Acquisition

- ♣ Chromosomal mutations
- ♣ Genetic transfer (ex: plasmids)

Section 11- Anthelmintic Drugs

1- DRUGS THAT ACT AGAINST NEMATODES

The medically important intestinal nematodes responsive to drug therapy include *Enterobius vermicularis* (pinworm), *Trichuris trichiura* (whipworm), *Ascaris lumbricoides* (roundworm),

- *Ancylostom.* and *Necator* species (hookworms), and *Strongyloides stercoralis* (threadworm).
- More than 1 billion persons worldwide are estimated to be infected by intestinal nematodes.

a- Albendazole

- **MOA:-** The action of albendazole is thought to involve inhibition of microtubule assembly.
- The drug is larvicidal in ascariasis, cystercercosis, hookworm, and hydatid disease and is ovicidal in ascariasis, ancylostomiasis, and trichuriasis.
- **Use:-**
 - Albendazole has a wide anthelmintic spectrum. It is a **primary drug for ascariasis, hookworm, pinworm, and whipworm** infections and an

alternative drug for treatment of threadworm infections, filariasis, and both visceral and cutaneous larva migrans.

- Albendazole is also used in hydatid disease and is active against the pork tapeworm in the larval stage (cysticercosis).
- Toxicity:- Few toxic effects during short courses of therapy (1–3 d). However, a reversible leukopenia, alopecia, and elevation of liver function enzymes can occur with more prolonged use.

b- **Diethylcarbamazine**

- **MOA**:- immobilizes microfilariae by an unknown mechanism, increasing their susceptibility to host defense mechanisms.
- **Use**:-Diethylcarbamazine is the **drug of choice for several filarial infections** including those caused by *Wucheria bancrofti* and *Brugia malayi* and for eye worm disease (loaloa).
- **Toxicity**:- Include headache, malaise, weakness and anorexia. In onchocerciasis, the reactions are more intense and include most of the symptoms described as well as hypotension, pyrexia, respiratory distress, and prostration.

c- **Ivermectin**

- **MOA**:-Intensifies γ -aminobutyric acid (GABA)-mediated neurotransmission in nematodes and causes immobilization of parasites, facilitating their removal by the RES.
- Selective toxicity results because in humans GABA is a neurotransmitter only in the CNS, and **ivermectin does not cross the blood-brain barrier**.
- **Use**:-Ivermectin is the drug of choice for onchocerciasis, cutaneous larva migrans, strongyloidiasis, and some forms of filariasis.
- **Toxicity** :- Single-dose oral treatment in onchocerciasis results in reactions to the dying worms, including fever, headache, dizziness, rashes, pruritus, tachycardia, hypotension, and pain in joints, muscles, and lymph glands. These symptoms are usually of short duration, and **most can be controlled with antihistamines and nonsteroidal anti-inflammatory drugs**. Avoid other drugs that enhance GABA activity. **Ivermectin should not be used in pregnancy**

d- **Mebendazole**

- **MOA**:- Acts by selectively inhibiting microtubule synthesis and glucose uptake in nematodes.

- **Use:-** Mebendazole is a **primary drug for treatment of ascariasis and for pinworm and whipworm infections**. Mebendazole has also been used as a backup drug in visceral larval migrans.
- **Toxicity:-** Usually limited to gastrointestinal irritation, but at high doses **agranulocytopenia** and alopecia have occurred. The drug is teratogenic in animals and therefore **contraindicated in pregnancy**.

e- Piperazine

- **MOA:-** paralyzes ascaris by acting as an agonist at GABA receptors. The paralyzed roundworms are expelled live by normal peristalsis.
- **Use:-** Piperazine is an alternative drug for ascariasis.
- **Toxicity :-** Mild gastrointestinal irritation is the most common side effect. **Piperazine should not be used in pregnant patients.**

f- Pyrantel Pamoate

- **MOA:-** Stimulates nicotinic receptors present at neuromuscular junctions of nematodes. Contraction of muscles occurs, followed by a depolarization-induced paralysis. The drug has no actions on flukes or tapeworms.
- **Use:-** It is a drug of choice for hookworm and roundworm infections and an alternative drug for pinworms. The drug is poorly absorbed when given orally.
- **Toxicity:-** Adverse effects are minor but include gastrointestinal distress, headache, and weakness.

g- Thiabendazole

- **MOA:-** Thiabendazole is a structural congener of mebendazole and has a similar action on microtubules.
- **Use :-** Because of its adverse effects, thiabendazole is an alternative drug in strongyloidiasis and trichinosis (adult worms).
- **Toxicity :-** it's toxic than the other drugs like ivermectin ,benzimidazole and its adverse effect includes gastrointestinal irritation, headache, dizziness, drowsiness, leukopenia, hematuria..etc we should avoid this drug in patients who had a renal disease and pregnancy.

2- DRUGS THAT ACT AGAINST TREMATODES

a- Prezquantel

- **MOA:-** Increases membrane permeability to calcium, then paralysis of of trematode and cestode muscles; followed by vacuolization and parasite death.
- **Use:-** It is the drug of choice in schistosomiasis (all species), clonorchiasis, and paragonimiasis and for infections caused by small and large intestinal flukes.
- **Toxicity :-** Common adverse effects include headache, dizziness and drowsiness, malaise, and less frequently, gastrointestinal irritation, skin rash, and fever.it has also neurological effects.in animal studies, the drug increased abortion rate.

b- Bithionol

- Bithionol is a codrug of choice (with triclabendazole) for treatment of fascioliasis (sheep liver fluke) and an alternative agent in paragonimiasis. Unknown **MOA**. Bithionol is orally effective and is eliminated in the urine.
- **Toxicity :-** Common adverse effects of bithionol include nausea and vomiting, diarrhea and abdominal cramps, dizziness, skin rash...etc

c- Metrifonate

- An organophosphate prodrug that is converted in the body to the cholinesterase inhibitor dichlorvos. These drug have many adverse effects but mainly it contradicts with pregnancy.

d- Oxaminquine

- Effective solely in *Schistosoma mansoni* infections (intestinal bilharziasis), acting on male immature forms and adult schistosomal forms. The drug causes paralysis of the worms, but its precise mechanism is unknown. Dizziness is a common adverse effect.

3- DRUGS THAT ACT AGAINST CESTODES(TAPEWORM)

- a- Niclosamid:** an alternative drug to praziquantel for infections caused by beef, pork, and fish tapeworm. Niclosamide is effective in the treatment of infections from small and large intestinal flukes.

Section 12- Anti-protozoan drugs

Disease caused by protozoans constitute worldwide health problem.

1- DRUGS OF MALARIA

- *P falciparum* and *P malariae* have only 1 cycle of liver cell invasion. The other species have a dormant hepatic stage responsible for recurrent infections and relapses.
- Primary **tissues chizonticides**(eg: primaquine) kill schizonts in the liver, whereas **blood schizonticides** (eg: chloroquine, quinine) kill these parasitic forms only in the erythrocyte.

Drugs used for the treatment of malaria as follows

a- Chloroquine

- Chloroquine is a 4-aminoquinoline derivative.
- **PK-** Given orally
 - Antacids may decrease oral absorption of the drug.
 - Excreted largely and unchanged in urine.

- **MOA:-** Chloroquine accumulates in the food vacuole of plasmodia and prevents polymerization of the hemoglobin breakdown product heme into hemozoin. Intracellular accumulation of heme is toxic to the parasite.
- **Resistance:-** Decreased intracellular accumulation via increased activity of membrane “pumps”.
- **Use:-** Chloroquine is the drug of choice for acute attacks of nonfalciparum and sensitive falciparum malaria and for chemoprophylaxis, except in regions where *P falciparum* is resistant. The drug is solely a blood schizonticide. Chloroquine and hydroxychloroquine are also used in autoimmune disorders, including rheumatoid arthritis.
- **Toxicity:-** at low dose, gastrointestinal irritation, skin rash and headaches whereas, in high dose symptoms like auditory impairment, retina damage..etc

b- Quinine

- **PK-** absorbed orally
 - Is metabolized before renal excretion.
- **MOA:** Quinine complexes with dsDNA to prevent strand separation, resulting in block of DNA replication and transcription to RNA.
- **Use :** - The main use of quinine is in *P falciparum* infections resistant to chloroquine in patients who can tolerate oral treatment. Quinine is commonly used with doxycycline or clindamycin to shorten the duration of therapy and limit toxicity.
- **Toxicity:-** commonly causes **cinchonism**, symptoms of which include gastrointestinal distress, headache, vertigo, blurred vision, and tinnitus. Severe overdose results in disturbances in cardiac conduction that resemble quinidine toxicity. **Blackwater fever** (intravascular hemolysis) is a rare and sometimes fatal complication in quinine-sensitized persons. Quinine is contraindicated in pregnancy.

c- Mefloquine- is synthetic 4-quinoline derivative.

- **PK-**It is given orally because of local irritation.
- **MOA:** -not known.
- **Use :-** first line drug given to prophylaxis.
- **Toxicity:-** common adverse effect includes gastrointestinal , distress, skin rash , headache and dizziness.at high doses, mefloquine has caused neurologic symptoms, seizures...

d- Primaquine - is synthetic 8-aminoquinoline.

- **PK-** Its given orally.
- **MOA :-**Primaquine forms quinoline- quinone metabolites, which are electron-transferring redox compounds that act as cellular oxidants. The drug is a tissue schizonticide and also limits malaria transmission by acting as a gametocide.

- **Use** :- the drug is also an alternative (daily) for primary prevention.
- **Toxicity** :- may cause gastrointestinal distress, pruritus, headaches, and methemoglobinemia. serious toxicity involves hemolysis in G6PD-deficient patients. Primaquine is contraindicated in pregnancy.

e- Antifolate drugs

- antifolate group includes pyrimethamine, proguanil, sulfadoxine, and dapsone.
 - **PK** - excreted in urine, partly in unchanged form.
 - proguanil the only drug with short half life(12-16h)
 - **MOA**:- Sulfonamides act as antimetabolites of PABA and block folic acid synthesis in certain protozoans by inhibiting dihydropteroate synthase. Proguanil (chloroguanide) is bioactivated to cycloguanil. Pyrimethamine and cycloguanil are selective inhibitors of protozoan dihydrofolate reductases.
 - **Use** :-Pyrimethamine with sulfadoxine in fixed combination (Fansidar) is used in the treatment of chloroquine-resistant forms of this species, although the onset of activity is slow. Proguanil with atovaquone in fixed combination (Malarone) can be used (daily) for chemoprophylaxis of chloroquine-resistant malaria and is also protective against mefloquine resistant falciparum strains.
 - **Toxicity** :- include skin rashes, gastrointestinal distress, hemolysis, kidney damage and also cause folic acid deficiency when used in high doses.

Other Anti-malarial drugs such as:-

- **Doxycycline**- is chemoprophylactic (taken daily) for travelers to geographical areas with multidrug-resistant *P. falciparum*.
- **Amodiaquine**- This drug has been widely used to treat malaria in many countries because of its low cost. It is also used in fixed combination with artenimol.
- **Atovaquone**-This quinine derivative, a component of Malarone (proguanil), appears to disrupt mitochondrial electron transport in protozoa. Malarone is effective for both chemoprophylaxis (taken daily) and treatment of falciparum malaria. Abdominal pain and gastrointestinal effects occur at the higher doses needed for treatment. Atovaquone is an alternative treatment for *P. jirovecii* infection.
- **Halofantrine**-Although its mechanism of action is unknown, this drug is active against erythrocytic. Halofantrine is not used for chemoprophylaxis because of its potential for quinidine-like cardiotoxicity and embryo toxicity.
- **Artesunate, artemether, dihydroartemisinin**-These artemisinin derivatives are metabolized in the food vacuole of the parasite forming toxic free radicals. The artemisinins are the only drugs reliably effective against quinine-resistant strains.

Adverse effect include also nausea, vomiting, and also abortion do not appear to be elevated in women treated with artemisinins during pregnancy.

2- DRUGS FOR AMEBIASIS

- Tissue amebicides(**chloroquine, emetines, metronidazole, tindidazole**) act on organisms in the bowel wall and the liver; luminal amebicides (**diloxanidefuroate, iodoquinol, paromomycin**) act only in the lumen of the bowel.
- The choice of a drug depends on the form of amebiasis.
- a- **Diloxanidefuroate**:- This drug is commonly used as the sole agent for the treatment of asymptomatic amebiasis and is also useful in mild intestinal disease when used with other drugs.
- b- **Emetines**:- These drugs are used parenterally (subcutaneously or intramuscularly) as backup drugs for treatment of severe intestinal or hepatic amebiasis together with a luminal agent in hospitalized patients. It may cause toxicity includes muscle weakness, cardiac dysfunction. The drugs are restricted to use in severe amebiasis when metronidazole cannot be used.
- c- **Idoquinol**:- is an orally active luminal amebicide used as an alternative to diloxanide for mild-to-severe intestinal infections. Adverse effects such as gastrointestinal effects.
- d- **Metronidazole and Tinidazole**:- are effective orally and distributed widely to tissues. The half-life of metronidazole is 6–8 h, and that of tinidazole 12–14 h respectively. Elimination of the drugs require hepatic metabolism.
 - **MOA**:- Metronidazole undergoes a reductive bioactivation of its nitro group by ferredoxin to form reactive cytotoxic products.
 - **Use**:- Metronidazole or tinidazole is the drug of choice in severe intestinal wall disease and in hepatic abscess and other extraintestinal amebic disease. The duration of treatment required with metronidazole is longer than with tinidazole. metronidazole is also used for treatment of giardiasis and trichomoniasis.
 - **Toxicity**:- Adverse effects of metronidazole include gastrointestinal irritation (it is best taken with meals), headache, paresthesias, and dark coloration of urine.
- e- **Paromomycin**:- used as a luminal amebicide and may be superior to diloxanide in asymptomatic infection may also have some efficacy against cryptosporidiosis in the AIDS patient.
- f- **Nitazoxanide**:- appears to have activity against metronidazole-resistant protozoal strains.

3- DRUGS FOR PNEUMOCYSTOSIS & TOXOPLASMOSIS

a- Pentamidine:- Unknown MOA.

- **Use**- Aerosol pentamidine (once monthly) can be used in primary and secondary prophylaxis, although oral trimethoprim-sulfamethoxazole (TMP-SMZ) is usually preferred.
- **Toxicity** :- Severe adverse effects follow parenteral use, including respiratory stimulation followed by depression, hypotension resulting from peripheral vasodilation...etc

b- TMP-SMZ(cotrimexazole):- is the first choice in prophylaxis and treatment of pneumocystis pneumonia (PCP). Prophylaxis in AIDS patients is recommended when the CD4 count drops below 200 cells/ μ L.

- **Toxicity**:- Adverse effects from TMP-SMZ occur in up to 50% of AIDS patients. Toxicity includes gastrointestinal distress, rash, fever, neutropenia, and thrombocytopenia.

c- Antifols: Pyrimethamine and Sulfonamides:-For treatment of active toxoplasmosis, the drug combination is given daily for 3–4 wk, with folinic acid to offset hematologic toxicity.

- **Toxicity** :- High doses of pyrimethamine plus sulfadiazine are associated with gastric irritation, glossitis...etc

d- Atovaquone:- Used orally, it is poorly absorbed and should be given with food to maximize bioavailability. Most of the drug is eliminated in the feces in unchanged form.

- **Toxicity**:- Atovaquone is approved for use in mild to moderate pneumocystis pneumonia. It is less effective than TMP-SMZ or pentamidine but is better tolerated. this drug should not be given to patients with cardiac defects, seizures and psychiatric disorders.

4- DRUGS FOR TRYPANOSOMIASIS

a- Pentamidine

- Pentamidine is commonly used in the hemolymphatic stages of disease caused by *Trypanosoma gambiense* and *T rhodesiense*. Because it does not cross the blood-brain barrier, pentamidine is not used in later stages of trypanosomiasis.

b- Melarsoprol

- Because it enters the CNS, melarsoprol is the drug of choice in African sleeping sickness.

- However, treatment failures do occur, possibly because of resistance. Melarsoprol is given parenterally because it causes gastrointestinal irritation; it may also cause a reactive encephalopathy that can be fatal.
- c- **Nifurtimox**:-Nifurtimox is the drug of choice in American trypanosomiasis, an alternative agent in African forms of the disease, and has also been effective in mucocutaneous leishmaniasis. It may cause gastrointestinal irritation and CNS effects.
- d- **Suramin**:- This polyanionic compound is a drug of choice for the early hemolympathic stages of African trypanosomiasis (before CNS involvement). It is also an alternative to ivermectin in the treatment of onchocerciasis.
- e- **Eflornithine**:- This agent, a suicide substrate of ornithine decarboxylase, is effective in some forms of African trypanosomiasis. It is available for both oral and intravenous use and penetrates into the CNS. It causes gastrointestinal irritation and hematotoxicity; seizures have occurred in overdose.

5- DRUGS FOR LEISHMANIASIS

- a- **Sodium stibogluconate** (pentavalent antimony): the primary drug in all forms of the disease, appears to kill the parasite by inhibition of glycolysis or effects on nucleic acid metabolism.

Section 13- Antimicrobials

1- ANTIBACTERIAL

Antibiotics- substance derived from one living organism which may kill or suppress the growth of another (anti-biosis)

Antimicrobial- antibiotic + semi synthetic + synthetic drugs

“Selective toxicity”- depending on the existence of exploitable biochemical differences between the invading organism and the host. Eg- Cell wall and synthesis of folate is found in bacteria not humans.

Bactericidal- kill the bacteria, Bacteriostatic- stop their growth so our own immune system kill the MO, so bacteriostatic drugs shouldn't be given to immunocompromised individuals.

- Classified based on **MOA**:

Cell wall synthesis inhibitor - interfere with the transpeptidation reaction of bacterial cell wall synthesis	Beta lactam products (Penicillin, Cephalosporin, Monobactam, Carbapenem) , Bacitracin, Vancomycin, Cycloserine
Cell membrane function inhibitor	polymixins, polyene antifungal
Protein synthesis inhibitors	amino glycosides, tetracycline's, chloramphenicol, macrolides, clindamycin, spectinomycin, mupirocin
Nucleic acid synthesis inhibitors	rifampin, quinolones, nitroimidazoles, nitrofurantoin
Energy metabolism modifier	sulfonamides, trimethoprim, dapsone

Cell wall synthesis inhibitor

➤ Penicillin- Bactericidal

1-Natural penicillin narrow spectrum	2-B-lactamase resistant very narrow spectrum
Penicillin G- acid labile, given IV, lipid insoluble (potent than Pen V) Penicillin- acid stable. Oral Use- for Gram positive Bac. B-lactamase sensitive	naficillin, methicillin, cloxacillin, dicloxacillin, flucloxacillin ,oxacillin food interferes with absorption Oral except nafcillin and methicillin Use- for staph infection(narrow spect.) Resistance- MERSA
3-Broad spectrum	4-Extended spectrum
Ampicillin, bacampicillin, amoxicillin --Don't give with food (except amoxicillin) Use- same as natural pens + G -ve bac (H.inf, E.coli,) B-lactamase sensitive	carbenicillin,ticarcillin,azlocillin, mezlocillin, piperacillin Parenteral Use- same as broad spec+ pseudomonas and klebsiella B-lactamase sensitive

B-lactamase inhibitor- clavulanic acid, sulbactam, tazobactam

Amoxicillin plus clavulanic acid= Augmentin, Carbenicillin plus clavulanic acid= Timentin

PK- renal excretion except nafcillin and oxacillin (lipid soluble- excreted through bile)

Adverse Effect- Hypersensitivity (tx- adrenaline), GI upset

Resistance of Beta lactams- inactivation by betalactamase, modification of target PBPs, impaired penetration of drug to target PBPs and efflux. Altered target PBPs is the case in methicillin resistant strains and penicillin resistant strains.

- **Cephalosporin-** similar to pen, but more stable to many b-lactamases, but susceptible to extended spectrum b-lactamases (E.coli, klebsiella)

1 st generation	Cefadroxil, cefazolin, cephalexin , cephalothin , cephapirin , cepharidine Very active against G +ve
2 nd generation	Cefaclor, cefamandole, cefonicid, cefuroxime, cefprozil, ceforanide, loracarbef Less activity against G +ve but extended G –ve function Ineffective against enterococcus or P.aerogenosa
3 rd generation	Expanded G –ve coverage than 2 nd generation Ceftazidime, cefoperazone- for P.aerogenosa
4 th generation	Extended spectrum than 3 rd generation- P.aerogenosa, enterococcus Increased stability from hydrolysis by B-lactamase
Not yet decided	Ceftaroline fosamil- For MERSA
	PK- renal except cefoperazone and ceftriaxone

-Half-life ranges from 0.5-8 hours

- Cross-allergenicity with penicillin (5-10%)

Adverse effect- hypersensitivity reaction, toxicity local irritation (IM), thrombophlebitis (IV), disulfiram like reaction (avoid alcohol)

Others

Monobactams Prototype- aztreonam, penetrate CSF	No activity against gram +ve bac or anaerobes Use- Limited to aerobic gram-negative rods (including P aeruginosa), Suitable for penicillin allergic patients stable to many beta-lactamases
Carbapenems imipenem, meropenem,	-instable at gastric PH (not given orally), distribute to CSF -B-lactamase resistant -imipenem- inactivated by dehydropeptidase (DHP) (give DHP inhibitor- Cilastatin) -reserve drugs for mixed aerobic and anaerobic infection. -most potent of all lactams (kaplan) -AE- include nausea, vomiting, diarrhea skin rashes; cross-allergenicity with pen. -imipenem- CNS effect- seizure
Glycopeptide	They also damage cell membrane

antibiotics- vancomycin,	-active only against gram-positive bacteria -bind to D-ala-d-ala muramyl pentapeptide and hinder transglycosylation -PK- IV except for C.difficile Adverse effects include <i>red man or red neck (TIHSR)</i> syndrome (prevented by antihistamine drug), chills, fever, ototoxicity and nephrotoxicity at high doses; slow infusion essential
Daptomycin	-Lipopeptide antibiotic obtained from soil microorganisms -Spectrum of activity similar to vancomycin, also for vancomycin-resistant strains of S aureus and enterococci -MOA- bind to the CM via calcium-dependent insertion of its lipid tail—results in depolarization of cell membrane with potassium efflux and rapid cell death - AE - allergic pneumonitis, myopathy (creatine phosphokinase level should be monitored)
Bacitracin	-active against gram-positive microorganisms -highly nephrotoxic when administered systemically and is only used topically . -indicated for the suppression of mixed bacterial flora in surface lesions of the skin, in wounds, or on mucous membranes
Cycloserine	-used almost exclusively to treat tuberculosis caused by strains of M.tuberculosis resistant to first-line agents - AE -serious dose-related central nervous system toxicity with headaches, tremors, acute psychosis, and convulsions

Inhibitors of protein synthesis

-Act by binding to bacterial ribosomes and interfering with protein synthesis

-30s ribosomal protein- Aminoglycoside, tetracycline

-50s ribosomal protein- chloramphenicol, macrolides and clindamycin

Aminoglycosides (streptomycin, neomycin, kanamycin, amikacin, gentamicin,)	-Bacterocidal - misreading of the code on mRNA template (Irreversible) - require O ₂ dependant uptake- Un-aerobes are resistant -PK- parenteral - poorly absorbed orally, elimination by the kidney Use - against <i>aerobic gram negative enteric</i> bacteria, esp in bacteremia, sepsis or endocarditis. - streptomycin - for mycobacterial infec and nontuberculous infec. AE - ototoxicity, nephrotoxicity, neuromuscular blockade (dec Ach release- Neostigmine can be used as antidote)
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	Resistance- production of conjugating enzyme which increase their elimination
Spectinomycin	Structurally related to aminoglycoside but with no amino sugars Bacteriostatic Solely used as an alternative treatment for penicillin resistant or allergic gonorrhea patients
Tetracycline	Bacteriostatic - block the binding of aminoacyl-tRNA to the acceptor site on the mRNA-ribosome complex Broad spectrum antibiotics Absorption impaired by presence of food, particularly divalent or trivalent cations(Ca^{2+}, Mg^{2+}, Fe^{2+}, Al^{3+})(except minocycline and doxycycline) Elimination by kidney (except- doxycycline) Use- declined lately. Used for M.pneumoniae, Chlamydia, rickettsiae AE-gastrointestinal adverse effects due to local irritation of intestinal tract -Readily bind to calcium deposited in newly formed teeth- CI in pregnancy and children < 8 yrs of age -Liver toxicity, kidney toxicity, photosensitization Resistance- efflux pumping
Macrolide Prototype-erythromycin Semisynthetic (clarithromycin, azithromycin)	Bacteriostatic or bactericidal depending on the MO Erythromycin- destroyed by acidic PH Use- for G +ve cocci Erythromycin- drug of choice in corynebacterial infections chlamydial infections and for community acquired pneumonia Inhibit cytochrome P450 except azithromycin SE- GI distress, cholestatic hepatitis, reversible deafness at high dose
Chloramphenicol	Bacteriostatic Good tissue perfusion – CSF Use- for sepsis and meningitis AE- GI disturbance, aplastic anemia(dose dependent), Newborn infants- lack effective glucuronic acid conjugation mechanism- gray baby syndrome Inhibits cytochrome p450
Clindamycin	Active against G+ aerobic cocci and anaerobic organisms Concentrate in bone- clinical value in osteomyelitis caused by

	S.aureus AE- pseudomembranous colitis
Oxazolidinones/Linezolid	

Anti- metabolites

Sulfonamides	-Bacteriostatic -structurally related to PABA, and enter in place of PABA -inhibit folate synthesis by inhibiting by inhibing dihydropteroate synthase -could be oral absorbable, nonabsorbable or topical Sulfamethoxazole + trimethoprim- cotrimexazole- for pneumocystis carini Pyrimethamine+ sulfadiazine- for T. gondi AE- HSR, G6PD deficiency hemolysis, phototoxicity -may precipitate in urine(neutral or acid), producing crystaluria, hematuria, or even obstruction- treated by admin. of sodium bicarbonate to alkalinize urine and fluids to increase urine flow. -displace bilirubin from plasma albumin of new born infants causing an encephalopathy called kernicterus; CI- pregnant women near term -increase the actions of oral anticoagulants -Trimethoprim produces megaloblastic anemia, leukopenia,and granulocytopenia
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DNA gyrase inhibitor

Quinolones/fluoroquinolones (nalidixic acid, norfloxacin, ciprofloxacin,levofloxacin, gatifloxacin)	-all fluorinated except nalidixic acid -MOA- Quinolones block bacterial DNA synthesis by inhibiting bacterial topoisomerase II (DNA gyrase) and topoisomerase IV -Use - Nalidixic acid used for UTI only/low syst.conc.
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2- ANTIVIRAL

a- Drugs for herpes and varicella zoster

-oral nucleoside analogs- acyclovir, valacyclovir, famciclovir

-they all have same MOA, acyclovir is the only one given IV.

	MOA	PK	indication	AE
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Acyclovir- acyclic guanosine derivative	-The drug is activated to form acyclovir triphosphate -compete with deoxyGTP for viral DNA polymerase, and bind to the DNA template as an irreversible complex; -chain termination following incorporation into the viral DNA	-topical, oral, and IV routes. -Renal excretion Resistance - changes in viral DNA polymerase or -Absence of thymidine kinase, enzyme involved in the initial viral-specific phosphorylation of acyclovir.	-oral-mucocutaneous and genital herpes and prophylaxis -IV- for severe herpes -more potent against HSV than VZV.	-well tolerated -Oral drug-GI distress and headache. -Parenteral-delirium, tremor, seizures, hypotension, and nephrotoxicity.
Valacyclovir	L-valyl ester of acyclovir Converted to acyclovir after oral admin.	-long duration of action than acyclovir	-for genital herpes, 1-day tx of orolabial herpes	-nausea, rash rarely - GI intolerance and thrombotic microangiopathy in AIDS pts.
Famciclovir	-prodrug, converted to penciclovir. -same MOA as acyclovir except that it doesn't cause chain termination	-Oral -renal excretion	-for HSV, VZV, EBV, HBV	-well tolerated -headache, diarrhea and nausea
Penciclovir	-guanosine analogue	For topical use	- for tx of recurrent herpes labialis in immunocompetent indiv.	-uncommon
Docosanol	Inhibit fusion between Plasma mem and HSV envelope	Topical	Herpes labialis	
Trifluridine	Inhibit viral DNA synt.	Topical	Herpes keratitis and acyclovir resistant HSV	
Foscarnet		-IV -renal excretion	Acyclovir resistant HSV and VZV	

b- Drugs for CMV

	<u>MOA</u>	<u>PK</u>	<u>Indication</u>	<u>AE</u>
Ganciclovir-	-inhibits DNA	-IV, orally or intraocular	-CMV	Myelosuppression

acyclic guanosine analog	polymerases of cytomegalovirus (CMV), and HSV and causes chain termination after being triphosphorylated -Valganciclovir – a prodrug of gan-ciclovir, has high oral bioavailability	- renal elimination in direct proportion to creatinine clearance.	-An intraocular implant form of ganciclovir can be used in CMV	esp after IV -nausea, diarrhea, retinal detachment in in CMV pts -CNS toxicity
Foscarnet	Inhibit viral DNA and RNA polymerase, HIV reverse transcriptase without phosphorylation	-IV only -30% maybe deposited in Bone marrow -renal excretion proportional to creatinine clearance	-CMV retinitis treatment	Renal impairment Hyper-or hypocalcemia -saline preloading to prevent nephrotoxicity
Cidofovir-cytosine nucleotide analogue	-phosphorylation independent of viral enzyme	IV- must be admin, with probenecid to decrease nephrotoxicity	For HSV, VZV, EBV, HHV-6, KSHV	Dose dependent nephrotoxicity Neutropenia

c- Antiretroviral therapy

NRTIs

- **MOA**-Prodrugs
- Competitively inhibit binding of natural nucleotides to dNTP-binding site of RT
- Chain terminators
- **AE**- cause lactic acidemia and sever hepatomegaly with steatosis

NNRTIs

- **MOA**-Bind to sites on RT different from the binding sites of NRTIs.
- Don't require phosphorylation, don't compete with nucleoside triphosphates.
- No cross-resistance with NRTIs.

PIs-

- Used mainly in combination with NRTIs as component of HAART.
- All are substrates and inhibitory of CYP3A4
- **AE**- disorders CHO and lipid met (**Buffalo hump, gynecomastia, truncal obesity, facial and peripheral lipodystrophy**)

	Class of agent	PK	AE
<u>Maraciroc</u>	Entry inhibitor	-Oral, good tissue penetration	
<u>Enfuvirtide</u>	Fusion inhibitor	Subcutaneous administration	Increased incidence of bacterial pneumonia has been reported
<u>Zidovudine (AZT)</u>	Prototype NRTIs	Active orally, CNS	Primary toxicity- Hematotoxicity, finger nail discoloration
<u>Abacavir (ABC)</u>	NRTI Guanosine analogue	oral	Hypersensitivity
Lamuvudine(3 TC)	NRTI	-80% oral bioavailability -also for HBV	One of the safest NRTI
<u>Tenofovir</u>	Nucleotide not nucleoside (NRTI)	Also has activity against HBV	
<u>Stavudine (d4t)</u>	NRTI		Due to its sever hepatic steatosis and lipodystrophy, d4T has phase out in the market
<u>Efavirenz,</u>	NNRTI <i>**a drug once thought to be teratogenic is proved not to be teratogenic and is used both in pregnant and nonpregnant women.</i>	Long half-life - once daily regimen -Fatty food enhances its bioavailability met by cyt P450	CNS dysfunction- insomnia and dysphoric
<u>Nevirapine</u>	NNRTI	Good oral bioavailability, penetrate most tissues- CNS Given in combination regimens	rash(HSR), Stevens-Johnson syndrome
<u>Delavirdine</u>	Prototype NNRTI		Drug interactions is the major problem ADE-skin rash

			Teratogenic in animals- avoid in pregnancy
<u>Etravirine</u>	Newest NNRTI	May be effective against strains resistant to the others	
<u>Raltegravir pyrimidine derivative</u>	Integrase inhibitor		
<u>Atazanavir</u>	PIs	Once daily dosing Absorption require acidic medium	-not associated with dyslipidemias or fat deposition unlike other PIs
<u>Ritonavir</u>	PIs	Good oral bioavailability, taken with food	GI irritation and bitter taste
<u>Lopinavir/ritonavir</u>	PIs	subtherapeutic dose of ritonavir acts as a pharmacokinetic enhancer by inhibiting the CYP3A4-mediated metabolism of lopinavir.	

population	Preferred first line	Alternative first line
Adults	TDF+ 3TC+ EFV	AZT+ 3TC+ EFV AZT+3TC+ NVP TDF+ 3TC+ NVP
Adolescents(10-19) > 35kg	TDF+ 3TC + EFV	ABC+ 3TC + EFV
Children 3-10 years and adolescents < 35kg	ABC+ 3TC+ EFV AZT+ 3TC+ EFV	TDF+ 3TC+ NVP
Children less than 3 years	ABC + 3TC + LPV/r AZT+ 3TC+ LPV/r	ABC+ 3TC + NVP AZT+ 3TC + NVP

3- Drugs for hepatitis

- **For hepatitis B-** Interferon alfa-2b, Peginterferon alfa-2a/-2b, Entecavir, Adefovir, Telbivudine Lamivudine/ Emtricitabine, Tenofovir .
- **For hepatitis C-** Ribavirin, Pegylated interferon

4- Anti-influenza agents

- -amantadine and rimantadine, zanamivir and oseltamivir.

3- Antifungal agents

a- Systemic antifungal drugs for systemic infections

	MOA	GENERAL	Adverse effect and toxicity
Amphotericin B	- interact hydrophobically with ergosterol in the fungal CM forming a pore- Fungicidal	-produced by <i>Streptomyces nodosus</i> -poorly absorbed in GIT -IV- for systemic disease -intrathecal for meningitis -topical for cutaneous	-infusion related toxicity -cumulative toxicity -renal damage -anemia -Intrathecal- seizure and a chemical arachnoiditis may occur.
Flucytosine	- taken up by fungal cells via the enzyme cytosine permease. It is converted IC first to 5-FU and then to FdUMP and FUTP, which inhibit DNA and RNA synthesis, respectively.	-Combined with amphotericin for severe infections such as cryptococcal meningitis	-infrequent: anaemia, neutropenia, thrombocytopenia & alopecia; reversed when therapy ceases
Azoles: based on number of nitrogen they could be Imidazole (Ketoconazole, miconazole, clotrimazole) or triazoles (Itraconazole, fluconazole, voriconazole, and posaconazole)	- MOA - interferes with synthesis of ergosterol by inhibition of fungal cytochrome P450 enzymes -The selective toxicity of azole drugs results from their greater affinity for fungal than for human cytochrome P450 enzymes. - all have the same MOA		
➤ Ketoconazole	. due to toxicity, exclusively	-Imidazoles exhibit a	-GI effects

	used topically	lesser degree of selectivity than the triazole - less specificity, therefore have more adverse effects. Penetrates poorly into CSF -	-hepatotoxicity -endocrine effect(gynecomastia)
➤ Itraconazole	<u>Use</u> - nonmeningeal histoplasmosis, used extensively in the treatment of dermatophytoses and onychomycosis	-absorbed orally but needs acid for dissolution -Penetrates poor into CSF.	-lacks endocrine effect because it doesn't inhibit cyt P450 responsible for synthesis of androgens.
➤ Fluconazole	Use -treatment and secondary prophylaxis of cryptococcal meningitis -treatment of mucocutaneous candidiasis	-good oral absorption -has the least effect of all the azoles on hepatic microsomal enzymes.	
Echinocandins	-newest antifungals -act at the level of the fungal cell wall by inhibiting the synthesis of $\beta(1-3)$ -glucan	-only given IV	

1- Oral treatment for mucocutaneous infections

	MOA	General	Adverse effects
Griseofulvin	-Derived from penicillium family -MOA at the cellular level is unclear, but it is deposited in newly forming skin where it binds to keratin, protecting the skin from new infection.	-Its only use is in the systemic treatment of dermatophytosis	-allergic syndrome much like serum sickness, hepatitis, and drug interactions with warfarin and Phenobarbital
Terbinafine	-Like the azole drugs, it interferes with ergosterol biosynthesis, but rather than	-It is used in the treatment of dermatophytoses, especially onychomycosis	-rare, consisting primarily of gastrointestinal upset

	interacting with the P450 system, it inhibits the fungal enzyme squalene epoxidase		and headache.
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2- Topical antifungal therapy

Topical azoles: Clotrimazole, miconazole	-for vulvovaginal candidiasis -In cream form, useful for dermatophytic infections, - Absorption is negligible, and adverse effects are rare. Topical and shampoo forms of ketoconazole are also available and useful in the treatment of seborrheic dermatitis and pityriasis versicolor.	Griseofulvin	-anti-dematophyte -interferes with mitosis (fungistatic)
Nystatin	-too toxic for parenteral use -active against most Candida sp and is most commonly used for suppression of local candidal infections -AE- bitter taste	TOPICAL ALLYLAMINES Terbinafine and naftifine	-Both are effective for treatment of tinea cruris and tinea corporis

4- Anti TB

First line

1- Isoniazid (INH)

- Most active antiTB drug.
- Bactericidal for rapid growers and bacteriostatic for slow growers.
- Effective against both intra- and extracellular bacilli.
- **MAO**
 - Cellwall(mycolic acid) synthesis inhibitor.
 - Pro drug activated by mycobacterial catalase.
- **Resistance** is developed by deletion of the catalase (katG gene).
- **PK- Absorption:** oral or parenteral route.
 - **Excretion:** by enzymatic acetylation
- Isoniazid is under genetic control, there are 2 groups of people.

- **Fast and slow acetylators**
 - **AE-** Related to dosage and duration of administration.
 - Immunologic reactions:
 - Fever
 - Skin rash
 - Drug induced systemic lupus erythematosus
 - Directtoxicity
 - **Age dependent hepatitis** (high AST & ALT) with histologic evidence of hepatocellular damage and necrosis.
 - Significantly associated with HIV co-infection.
 - Risk is higher in alcoholics, pregnancy and postpartum period.
3. Peripheralneuropathyin10-20% more likely in slow acetylators
 - Due to relative deficiency of pyridoxine.
 4. CNS toxicity, less common and may also respond to pyridoxine.
 5. Miscellaneous adverse effects:
 - Provocation of pyridoxine deficiency anemia, tinnitus and gastrointestinal discomfort.
 6. SLE (rare) in slow acetylators

2-Rifampin

- **MOA:** Binds to the β subunit of bacterial DNA–dependent RNA polymerase and thereby inhibits RNA synthesis (nucleic acid synthesis)

Clinical uses:

(1) TB

(2) Leprosy and other bacteria.

Adverse effect

- Harmless orange red color in all body excretions.
- Rash, cholestasisjaundice, flu like syndrome, **hepatitis**, nephritis

DDI

- Induces p450 isoforms.
- Induce microsomal drug metabolizing enzymes which will decrease the half-life of some other drugs like Anticoagulants, cyclosporine, anticonvulsants(phenytoin), contraceptives, methadone,PIs, NNRTIs.

3- Pyrazinamide

Weakly tuberculocidal.

- Active against intracellular bacilli.

- Pro drug Converted to pyrazinoic acid.

Adverse effects

1. Hepatotoxicity (major adverse effect).
2. **Hyperuricaemia** (it may provoke acute gouty arthritis).
3. Nausea, vomiting, & drug fever.

4- Ethambutol

MOA: inhibit synthesis of arabinogalactan (cell wall component)

- Suppresses growth of most INH and STM resistant
- Resistance develops very slowly and is due to mutations resulting in over expression of emb gene products or within the emb B structural gene

Adverse effect

- Most common and serious is optic neuritis -> loss of color vision.

Second line

Drugs	MOA	Indication	PK	AE
<u>Streptomycin</u>	Protein synthesis inhibitor	Bactericidal for extracellular bacilli	IV	Affects CN8 Nephrotoxicity (decrease Ach release)
Ethionamide	blocks synthesis of mycolic acids	For susceptible organisms.	Oral	
Capreomycin	peptide protein synthesis inhibitor	For streptomycin or amikacin resistant TB	IM	Nephrotoxicity Ototoxicity Local pain & sterile abscesses
Cycloserine	inhibiting alanine racemase(cell wall synthesis)	For resistant to first line agents.		CNS dysfunction(supplement P yridoxine 150mg/day) Peripheral neuropathy. Seizures, Tremors
Aminosalicyclic acid	folate synthesis antagonist	Exclusively for TB		
Fluoroquinolones	Inhibit DNA synthesis by inhibiting bacterial topoisomerase II and topoisomerase IV.	also for atypical mycobacteria		Nausea, vomiting, diarrhoea (most common) Headache, dizziness, insomnia, skin rash, photosensitivity. Damage growing

				cartilage and arthropathy. Tendinitis, tendon rupture.
Kanamycin		For streptomycin resistant		
Amikacin		For MDR and atypical TB	IV	
Linezolid		Last resort for MDR		BM depression Irreversible peripheral and optic neuropathy reported with prolonged use of drug
Rifabutin	derived from rifamycin. -cross-resistance with rifampin is virtually complete	Less potent inducer, rifabutin is indicated in for tx of Tb in patients with HIV infection who are receiving antiretroviral therapy with a PI or a NNRTI(eg, efavirenz), drugs that are cytochrome P450 substrates		

5- Anti-leprosy

-WHO recommends a multidrug therapy. This includes:

- **Dapsone (diaphenylsulfone, DDS), Rifampicin, Clofazimine, Ofloxacin** , and Minocycline constitute the backbone of the multidrug therapy (MDT) regimen.

-There are virtually no relapses

-No resistance of the bacillus to MDT has been detected;

	MOA	Clinical use	ADE
Dapsone	-like the sulfonamides, inhibits folate synthesis (PABA antagonist), Bacteriostatic	-Tuberculoid leprosy: with rifampin -Lepromatous leprosy: with rifampin and clofazimine -Prevention and treatment of pneumocystis jiroveci pneumonia in AIDS	-Haemolysis (in patients having G6PD deficiency). -Methemoglobinemia -GI intolerance -Fever -Pruritus and various rashes -Erythema nodosum

		patients. PK- Sulfones are well absorbed from the gut and widely distributed throughout body fluids and tissues.	leprosum (give corticosteroid or thalidomide)
Rifampicin			-A slight reddish coloration of urine, sweat, and tears
Clofazimine	-Unknown, but may involve DNA binding. -Clofazimine binds to DNA & inhibits template function. Its redox properties may lead to generation of cytotoxic oxygen radicals that are also toxic to the bacteria. -bactericidal	-Crosses placental barrier and is excreted into breast milk in significant amounts. -The drug does not cross the BBB.	-loss of appetite - Diarrhea -Nausea, vomiting -Dry skin and discoloration (from pink to brownish-black) -Gastrointestinal intolerance occasionally.(Eosinophilic enteritis)

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